

Technology Stack

Date	9 July 2026
Team ID	Not Assigned
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side <i>(e.g., React, Angular, HTML/CSS)</i>	<i>HTML, CSS, JavaScript (React.js optional)</i>	<i>Provides user-friendly interface for farmers to input data and view results; responsive and easy to use</i>

2	Backend / Server-Side <i>(e.g., Node.js, Python/Django, Java)</i>	<i>Python (Flask)</i>	<i>Efficient for ML integration, fast processing of input data, easy to connect with ML model</i>
3	Database / Data Storage <i>(e.g., MySQL, MongoDB, PostgreSQL)</i>	<i>CSV Dataset / MongoDB (optional)</i>	<i>Stores crop dataset; CSV used for ML training, MongoDB can store user inputs if needed</i>
4	Cloud / Hosting / Deployment <i>(e.g., AWS, Azure, Heroku, Docker)</i>	<i>Render / Vercel</i>	<i>Easy deployment of backend and frontend, cost-effective and beginner-friendly</i>
5	Version Control & CI/CD	<i>GitHub</i>	<i>Enables version control, collaboration, and project management</i>

	<i>(e.g., GitHub, GitLab, Jenkins)</i>		
6	Third-Party APIs / Other Tools <i>(e.g., Stripe, Twilio, Google Maps)</i>	<i>Weather API (optional), Scikit-learn</i>	<i>Weather API provides real-time data; Scikit-learn used for ML model training and prediction</i>